PAPER_885: GW Damping Erosion 66%

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-08 **Session:** 209 **Source:** Sessions 204-208 standalone module integration **Calculator:** GW-DampingErosion66PercentCalc (CP4 #469) **CVW:** v2.0.0 compliant

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Abstract

GW170817 gravitational wave damping via buoyancy erosion. h_damped = h_GR × (1 - D_erosion) where the combined damping fraction D = 66.7% (PAPER_008b constraint). Phase lag $\Delta\varphi$ = D × f_GW/f_orbit cycles. Demonstrates E^{-}(t) erosion engine applies to GW strain reduction.

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1. Core Equations

```
h_damped = h_GR × (1 - D)
D_erosion ≈ 0.667
Δφ = D·f_GW/f_orbit
```

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2. Parameters

Parameter	Default	Description
h_GR	1e-21	GR gravitational wave strain
f_GW	100 Hz	Gravitational wave frequency
f_orbit	50 Hz	Orbital frequency
D_erosion	0.667	Damping fraction

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3. UQFF Integration

This calculator integrates negative_et_erosion.py from Session 205 into the CP4 pipeline as class #469.

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